

3. Play Black Piano Keys

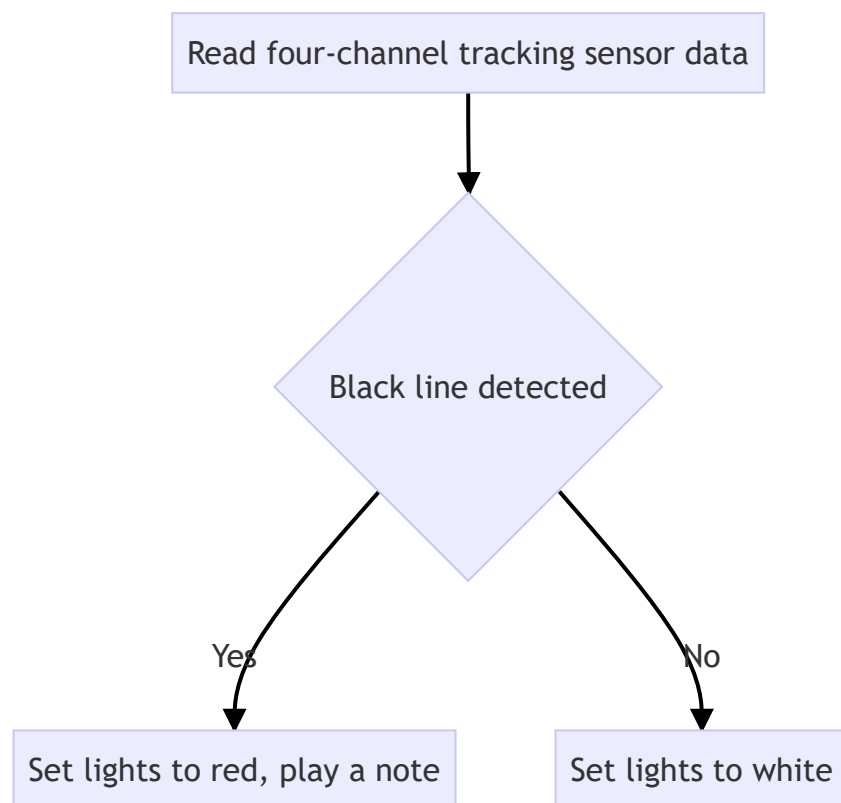
3. Play Black Piano Keys

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Function Description

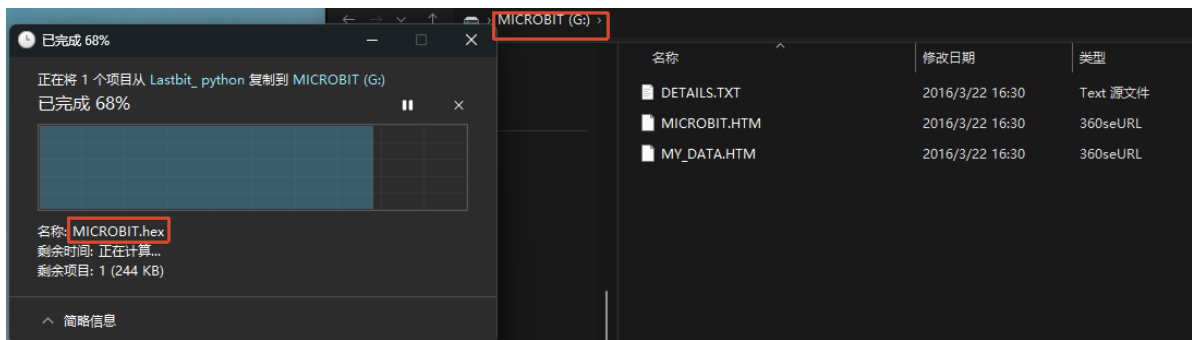
In this section, we will learn to use the line-tracking sensor to detect black lines, so that the car can "play" musical notes. When the car detects a black line, it will make sounds of different pitches, as if playing the piano.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



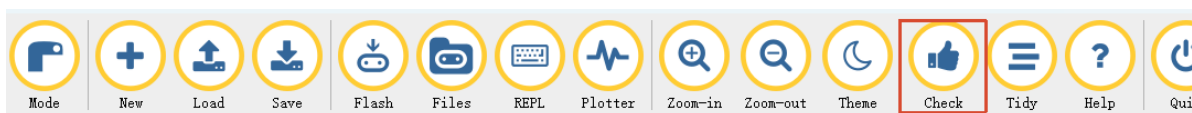
1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.
Click the **New** button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

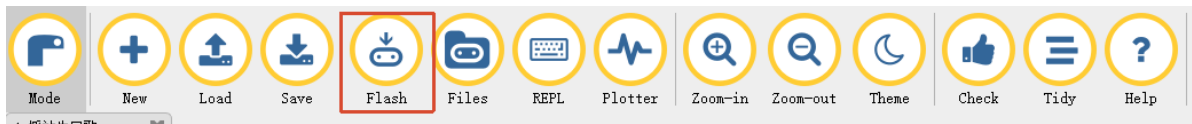
4. Now you can click **Check** to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows **Copied code onto micro:bit.** at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 3. Play Black Piano Keys
from microbit import *
from lastbit import LASTBIT
import music

Lastbit = LASTBIT()

# On the line or not
on_line = False
# Current note count
note_count = 0
# Last note count
last_note_count = 0

def black_line_detection():
    """Black line detection"""
    global on_line, note_count

    # Read track sensor data
    track = Lastbit.read_track()
    if track is None:
        return

    # Data conversion: 0 to 1, 1 to 0
    left = 1 if track[0] == 0 else 0
    mid_l = 1 if track[1] == 0 else 0
    mid_r = 1 if track[2] == 0 else 0
    right = 1 if track[3] == 0 else 0

    sum_val = left + mid_l + mid_r + right

    # Detect black line
    if sum_val > 0 and not on_line:
        Lastbit.set_headlight_both('RED')
        note_count += 1
        # Adjust speed and time if the battery is low and the car cannot get off
        the black line
        Lastbit.car_control('FORWARD', 60, 500)
        on_line = True

    # Leave black line
    if sum_val == 0:
        Lastbit.set_headlight_both('WHITE')
        Lastbit.car_control('FORWARD', 60)
        on_line = False

# Note frequencies
```

```

notes = [262, 294, 330, 349, 392, 440, 494]

display.show(Image.HEART)

# Initialize servos
Lastbit.set_all_servos(90)
sleep(200)
Lastbit.set_servo('S2', 150)

while True:
    # Play tone
    if note_count != last_note_count:
        if 1 <= note_count <= 7:
            music.pitch(notes[note_count - 1], 500)
        if note_count == 7:
            note_count = 0
        last_note_count = note_count

    # Detect black line
    black_line_detection()

    sleep(10)

```

Experimental Phenomenon

The car moves forward. **Due to mechanical errors in the motors, the tracks may slip during movement. These factors can cause the car to fail to move in a straight line even when the left and right motor speeds are set the same.**

When the four-channel tracking sensors detect a black line, they will light up red and play different notes.

When none of the sensors detect black, the white lights will be on.

Notes

Make sure the battery is fully charged and the four-channel tracking sensors have been adjusted to appropriate sensitivity via the potentiometers.